

Yelp Dataset Sentiment Analysis

Phase 2: Project Plan

Cole Hanrahan and Julia Sober
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GEORGETOWN UNIVERSITY

Project Motivation and Goals

- Star ratings on Yelp reviews are widely used but can be subjective and inconsistent.
- Predicting ratings from review text can help users gauge sentiment efficiently, businesses pinpoint improvement areas, and researchers better understand satisfaction drivers.
- **We are building on prior work in text-based sentiment and rating prediction, aiming to improve model performance and address gaps in contextual understanding and sentiment analysis.**

Insights from Literature Review

- Sentiment classification (positive vs. negative) achieves high accuracy (around 90%), but predicting specific star ratings (1–5) is much harder, with accuracy between 60–65%.
- Best traditional features are TF-IDF n-grams, and strong classical models include logistic regression and SVM.
- Deep models like CNNs and BERT offer nuanced understanding but struggle particularly with middle (2–3 star) reviews due to ambiguity and class imbalance.
- **Key gaps:** limited context understanding, dominance of 4/5-star reviews, and need for better model explainability.

Data Preparation and Baselines

- Filtered Yelp restaurant reviews, normalized text (lowercasing, punctuation and stopword removal), and derived auxiliary features like review length.
- Output: cleaned, balanced dataset ready for modeling and performance benchmarking.
- Built baselines with TF-IDF features and classic models (Logistic Regression, Naive Bayes, SVM, Random Forest, Ordinal Logistic Regression).
 - “Binary” = positive/negative/neutral
 - 1-5 RRP

Baseline Results (subsample n=500,000)

Features = Unigram + Bigram + TF_IDF w/ max_features = 10,000

Model	Classification Type	Accuracy (%)	Macro-F1
Logistic Regression	1-5 RRP	67.66	0.5969
Naive Bayes	1-5 RRP	64.15	0.5187
SVM	1-5 RRP	44.70	0.4147
Random Forest	1-5 RRP	51.96	0.2869
Ordinal Logistic Regression	1-5 RRP	60.92	0.5268
Logistic Regression	Pos. / Neu. / Neg.	87.43	0.7398
Naive Bayes	Pos. / Neu. / Neg.	84.45	0.6494
SVM	Pos. / Neu. / Neg.	60.63	0.5081
Random Forest	Pos. / Neu. / Neg.	75.83	0.4622
Ordinal Logistic Regression	Pos. / Neu. / Neg.	85.66	0.7133

What We Noticed (From Baselines)

- The Pattern:
 - **Positive vs Negative** reviews are *very easy* to separate (97% accuracy when we took out 3 star ratings).
 - **Neutral (3-star)** reviews are *ambiguous* — they overlap lexically and sentiment-wise with both extremes.
 - The multi-class (1–5) setup suffers because 3-star reviews blur the boundaries.
- The model can almost perfectly identify polarity, but it struggles to map polarity → precise star rating.
- This means the information bottleneck isn't “sentiment detection” — it's “rating calibration.”

Experimental Tests

1. Layered/Hierarchical classifier

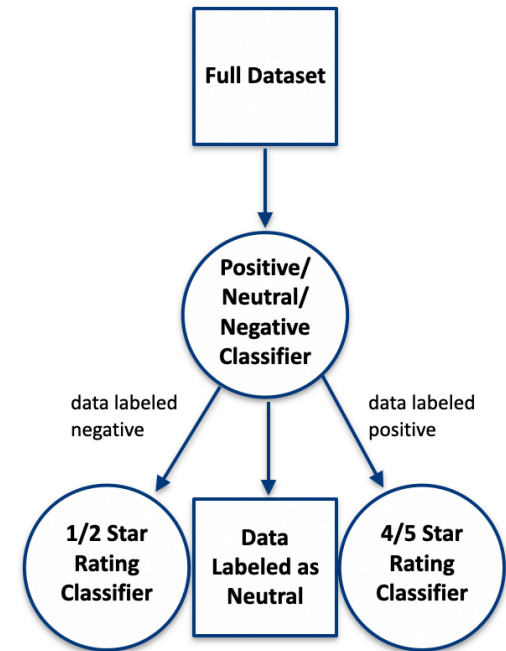
- TF-IDF + n-grams with Logistic Regression
- Stage 1: Sentiment classification (negative/neutral/positive)
- Stage 2a: Negative reviews → 1 vs 2 stars
- Stage 2b: Positive reviews → 4 vs 5 stars
- Neutral reviews → automatically assigned 3 stars
- Slightly better than baseline in some classes, worse in others, but limited improvement overall (< 0.02%)

2. Layered classifier with soft probabilities

- Passing probabilities from layer 1 classifier
- Used pre-trained sentence embeddings (s-BERT) instead of TF-IDF
- Worse accuracy for binary (layer 1) classifier than baseline
- Layer 2 classifier ended up predicting dominant class (5-star) for every rating

3. Simple MLP with sBERT

- Adam optimizer, 100 epochs, 1 hidden layer (128), ReLU activation, batch size = 64
- 62% test accuracy



Report accuracy after final classifiers finish

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=== Star Rating Classification Report ===
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	precision	recall	f1-score	support
1.0	0.000	0.000	0.000	6055
2.0	0.000	0.000	0.000	4186
3.0	0.000	0.000	0.000	5605
4.0	0.287	0.008	0.015	11892
5.0	0.445	0.994	0.615	22262
accuracy			0.444	50000
macro avg	0.147	0.200	0.126	50000
weighted avg	0.267	0.444	0.277	50000

Star Rating Accuracy: 0.4444

Our Proposal

- Best test accuracies:
 - Logistic Regression (Classifier) w/ TF-IDF & n-grams => 67%
 - Linear Regression (Regressor) w/ TF-IDF & n-grams => 68% (rounding up to nearest star)
- These accuracies are already 5-7 percentage points better than all of the papers we read (we trained on more data than papers).
- We can potentially improve these results even more with hyperparameter tuning.

Project Roadmap

9/16 – 10/03

10/04 - 10/31

11/01 – 11/10

11/11 – 12/08

Phase 1

Literature Review

Data Gathering

Cole & Julia

Cole & Julia

Phase 2

Data Cleaning

Json to csv files
Restaurant reviews only

State specific data frames

Preprocessing

Lowercasing
Negation Handling
Removing stopwords, punctuations, urls, numbers
Extract review length, capitalizations, exclamation points

Feature Extraction

TF-IDF features
Unigrams + bigrams

Phase 3

Baseline Modeling

Linear Regression
Logistic Regression

Cole

Naïve Bayes

SVM

Random Forest
Ordinal Logistic Regression

Julia

Advanced Modeling

Layered classifier + soft routing

Simple MLP + BERT

Multi-Task BERT

Julia

Hyperparameter tuning and model adjustments

Cole

Final Evaluations

Error Analysis

Per class performance

Evaluation & Comparison

Confusion matrices & metric tables

Cole & Julia